

HAT-P-36b

R-1472

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_210524 · created 2026-08-01T11:56:58+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459358.7457 +/- 0.004 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.066 +/- 0.025
Transit depth (Rp/Rs)^2	0.43 +/- 0.33 [%]
Orbital Inclination (inc)	89.67 +/- 2.8
Airmass coefficient 1 (a1)	1.104 +/- 0.016
Airmass coefficient 2 (a2)	-0.079 +/- 0.014
Scatter in the residuals of the lightcurve fit is	1.47 %
Best Comparison Star	#3 - [99, 71]
Optimal Method	PSF photometry
Transit Duration (day)	0.0907 +/- 0.0035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.066 ± 0.025 vs 0.126 (deviation 2.31σ)
Duration fit vs published	0.0907 d vs 0.0925 d (deviation 0.49σ)
Mid-time O-C	9.5 min
Depth SNR	2.5
Residual scatter	1.47 %

Light curve

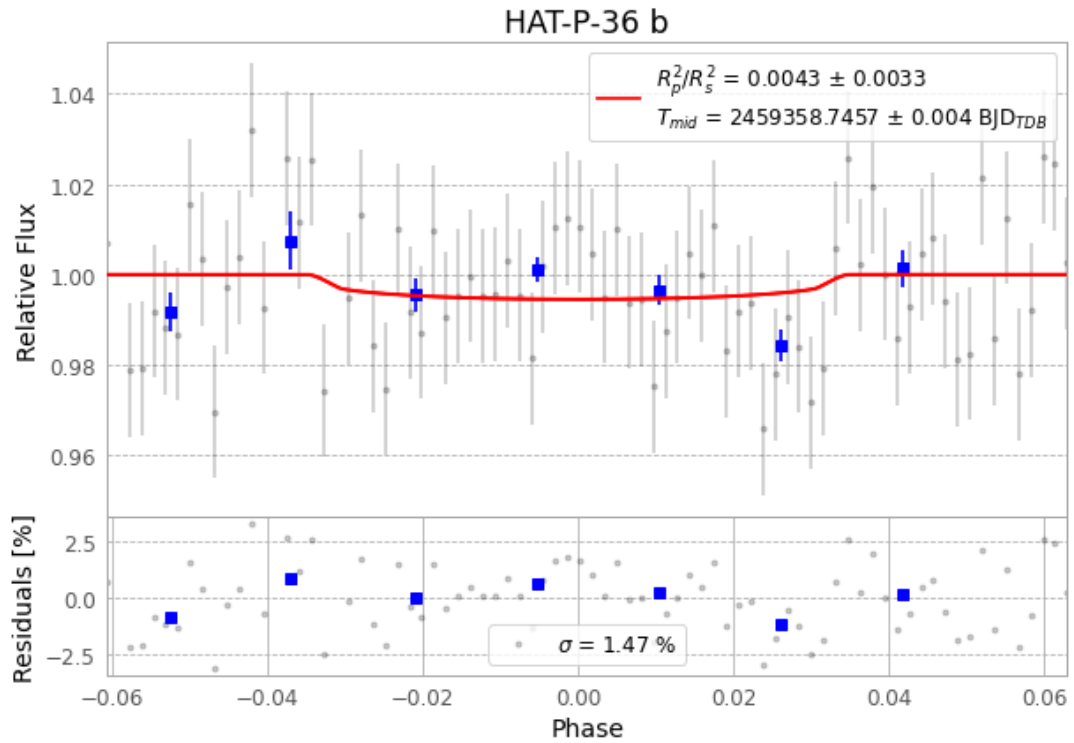


Figure 1 — FinalLightCurve_HAT-P-36 b_2021-05-23.png (SHA-256 fea044bd90f3b4f2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [402, 236], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5ffd7639af33d62a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

79 FITS frames (52 MB), HATP-36210524035443.FITS → HATP-36210524075112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-210524.log (fcb8c80eb8ab...), FinalLightCurve_HAT-P-36 b_2021-05-23.png (fea044bd90f3...), FinalLightCurve_HAT-P-36 b_2021-05-23.csv (72a74aa6c29a...)

Compute resources

Wall time 140.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 11:56Z.